

BSIT 2-1N

Activity #2 - Operators

General Instructions:

1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., `System.out.println(86);` without computation is not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use `System.out.println()` for displaying output.
8. Screenshot the code and the output.
9. Compile and run your program to verify correctness before submission.
10. Submission Format:
Section_LastName_FirstName_OperatorsActivity#2.pdf

Please answer the following questions:

PART I

- Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

CODE:

```

/**
 * @author Javier
 */
public class BitwiseActivity {
    public static void main(String[] args) {
        boolean a = true;
        boolean b = false;
        System.out.println(a | b);

        int numA = 8;
        int numB = 2;
        System.out.println(numA | numB);

        System.out.println(a & b);

        int numC = 7;
        int numD = 8;
        System.out.println(numC | numD);
    }
}

```

OUTPUT:

```

true
10
false
15

=== Code Execution Successful ===

```

- Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

SOURCE CODE:

```

/**
 * @author Javier
 */
public class TernaryActivity {
    public static void main(String[] args) {
        int x = 67;
        int y = 21;

        System.out.println((x > y) ? true : false);
        System.out.println((x < y) ? true : false);
        System.out.println((x == 67) ? 25 : 0);
        System.out.println((y == 67) ? 0 : 30);
    }
}

```

OUTPUT:

```

true
false
25
30

=== Code Execution Successful ===

```

- Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

SOURCE CODE:

```

/**
 * @author Javier
 */
public class AssignmentActivity {
    public static void main(String[] args) {
        int a = 67;
        a += 53;
        System.out.println("Addition: " + a);

        int b = 137;
        b -= 67;
        System.out.println("Subtraction: " + b);

        int c = 72;
        c *= 2;
        System.out.println("Multiplication: " + c);

        int d = 64;
        d /= 4;
        System.out.println("Division: " + d);
    }
}

```

OUTPUT:

```

Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16

=== Code Execution Successful ===

```

PART II: Program Requirements

- Use at least five (5) different operators from the following categories:
- Arithmetic operators (+, -, *, /, \%)
- Relational operators (>, <, >=, <=, ==, !=)
- Logical operators (&&, ||, !)
- Unary operators (++, --)
- Assignment operators (+=, -=, *=, /=)
- Bitwise operators (&, |, \wedge, <<, >>) The program must produce exactly six (6) outputs only.

- At least one output must include a tricky increment or decrement case, such as
 - `a++ ++a`
 - `--b++`
 - or any similar expression that modifies the variable within the same statement.
- Do NOT reuse
 - any variable values.
 - any variable names.
 - any logic patterns from Part I.
- The output must contain
 - At least two (2) boolean results.
 - At least four (4) numeric results.

SOURCE CODE:

```
* @author Javier
*/
public class AdvancedOperators {
    public static void main(String[] args) {
        int alpha = 5;
        int beta = 8;
        int omega = 50;

        int trickyResult = alpha++ + --alpha;
        System.out.println(trickyResult);

        boolean logicalRelational = (beta > 3) && (omega <= 100);
        System.out.println(logicalRelational);

        boolean logicalRelational2 = (omega == 40) || !(beta != 9);
        System.out.println(logicalRelational2);

        omega %= 7;
        System.out.println(omega);

        int delta = beta * 3 - 4;
        System.out.println(delta);

        int epsilon = 2 << 2;
        System.out.println(epsilon);
    }
}
```

OUTPUT:

```
10
true
false
1
20
8

=== Code Execution Successful ===
```